

UNIT 2

Working Methodology

Dust collection system is designed to remove particulates from the air produced during a production operation. This short definition is an extreme simplification of the ingenuity required to design and produce a means of capturing harmful contaminants.

The basic components of a dust collection system include a blower, dust filter, cleaning system, receptacle, ductwork, and means of collecting the particulate matter. The common types of dust collecting equipment include fabric filter baghouses, inertial separators - sometimes referred to as mechanical cyclones, cartridge collectors, wet scrubbers, and electrostatic precipitators. Baghouse dust collectors are the most commonly used since they have a 99% efficiency.

The types of pollutants removed differ according to the industry. Dust collector manufacturers design and develop equipment to specifically meet the needs of each environment condition.

Dust Collection System Works

Dust collection systems have ductwork to draw in the air, an air purifier, and receptacle. These basic elements are configured differently for each type of system.

Following are the **methods** of dust collection generally used in thermal power plants using coal as a fuel

(a) *The settling chamber*: In this method of dust collection the principle used is to decelerate below the free falling velocity of the particles in it.

(b) *Impingement separators*: These type of separators work on the principle of initial effect produced on the particles when sudden change of direction of gas flow is made. In this are the particles which move along with the gas stream, the large portion having finite mass and momentum gas when a sharp turn is provided but continue its original direction of motion and fall behind.

(c) *Cyclone separators*: In this system a whirling column of gas is produced by means of axial tangential and into the cylindrical body. As the cylindrical body is rotated due to centrifugal force wing on the particles, It makes them to move to the periphery of gas stream and in the runs the collide with the walls of cyclone well, thus separating them due to their mass Figure A represents a cyclone type dust collector. It consists of an axial type vane inlet. This spins the gas and dust into helix due to centrifugal forces. It moves along the periphery of the cylinder. Here nearly 5 to 10% of the gas containing most of the dust particles move to primary hopper section. Here larger particles set down due to gravity. Then gas goes to the bank of secondary cyclones, where smaller particles are removed.

(d) *Fabric filters*: In this system the gases are made to pass through woven cloth which filter out the dust from them. This method is complicated and involves combination of impaction, diffusion, thermal and electrostatic forces. The life of such filters depends upon operating temperature and surrounding conditions. They require Higher operational temperatures, so as to avoid, alkaline dust, water dew points and 2018 dew points. Terylene is best material for filters, but it is decomposed by hydrogen chloride. The next suitable material is glass fibre but it is very weak mechanically.

(e) *Electrostatic precipitators*: In works on the principle, that when gas consisting of dust particles is made to flow into a chamber, where dust particles are given an electric charge and on passing through the electric held the charged particles are deflected on to collecting plates due to the electric force acting on them.

A parallel-plate type precipitator which is most commonly used in thermal plants now-a-days is shown in Figure B. It consists of vertical plates as shown in Figure B. This also serves as collecting electrodes. In between them are hunged vertically discharged electrodes at equal distances. They can be made from round, square wire or from a strip. They are insulated from collecting electrodes and are connected to the negative terminal of high D.C. voltage. This type of arrangement helps in starting corona discharge, which produces electrons and thereby ionizes the gas molecules by collision. The dust particles on passing between the electrodes in the electric field are negatively charged by collision or diffusion and start moving towards the collecting electrodes in a very short time. They come in contact with the collector electrode. Their charge is neutralized and fall under gravity along the side of these electrodes. It is then collected by some mechanical means.

- **Dust Collection Ductwork**

The design of the ductwork may seem to be simple but has to be carefully considered to ensure proper performance of the system. The size of the pipe used depends on the tool size, air requirements, length of needed pipe, number of machines being serviced, and the types of particulates being extracted. The size of the ductwork changes throughout the system and collects the air drawn in by fans and collectors.

- **Fans or Blower**

Though a fan or blower may have a simple design, when installed in a dust collection system, several factors have to be considered. First of those factors is the volume of air that needs to be moved. This is measured in cubic feet per minute (CFM). The next consideration is the static pressure throughout the whole system. Other variables are the temperature, substances in the air, and the level of moisture.

The blower or fan is a critical element in a dust collection system since it is the mechanism that pulls the contaminated air into the ductwork away from the workplace and sends it to the filtration and cleaning systems. The basic types of blowers are centrifugal and axial. The centrifugal type has wheels in a housing, while the axial type has propellers.

- **Dust Filter Cleaning System**

As can be imagined, the filtration system for a dust collection system can become filled and clogged as particulates build up on the surface of the filter. There are a variety of methods used to clean filters some of which involve shutting down the system. For systems that have to be in continuous operation, this is not an option and during operation alternate methods have to be used.

An on-demand system has a controller with a pressure sensor for monitoring the static differential on the filter. The system measures the pressure differential between the clean air plenum and the dirty air plenum on the filter. When the levels are too high, it sends a signal to a diaphragm valve to release compressed air into the filter to remove the accumulated particles. This is referred to as a pulse jet baghouse or pulse jet dust collector and is the most common form of particulate air pollution control equipment. Other systems produce an alarm to advise operators that a significantly high pressure has been reached. Most systems, whether they are on demand or not, have control devices that make operators aware of a collection failure or drop in pressure across the filter.

Types

- **Baghouse Dust Collection Systems**

The most common form of dust collecting system is the baghouse. It is the most efficient system that uses a fan to create a vacuum to force contaminants through a filter in a bag. Baghouse methods are divided by how they clear their filters. The shaker method shakes the filter to remove the collected dust. The jet version uses a burst of air when a sensor detects the filter is full. The reverse air method forces fresh air through the filter. Some of the industries that use baghouse systems are powder coating companies, cement plants, paper manufacturers, and rubber recycling.

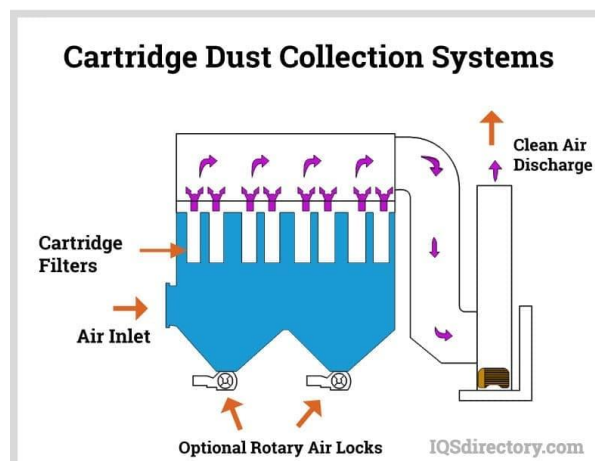
Dust cakes on the outside of the bags and is removed down into collection hoppers by a blast of compressed air or being shaken. They operate continuously to collect particles of any size. The design of the system allows for easy access for maintenance.

- **Cartridge Dust Collection Systems**

There are different forms of cartridge dust collection systems with some systems having the cartridges hung vertically while other systems have the cartridges placed horizontally. The design of the two systems is to meet the needs of different air filtering processes.

With a horizontal system, dirty air enters the cartridge at high velocity. The top row of cartridges have deflectors to protect the other cartridges and to ensure the cartridges do not get clogged. As the dirty air goes through the cartridges, it is deposited on the external part of the filter. In cyclical fashion, a burst of air is injected into the cartridge that creates a wave effect through the filter shaking the collected material into the hopper. The horizontal system allows for permanent and perfect cartridge alignment without yokes, sagging, or cracking. Gaskets are used to create a tight and secure seal along the housing and filters.

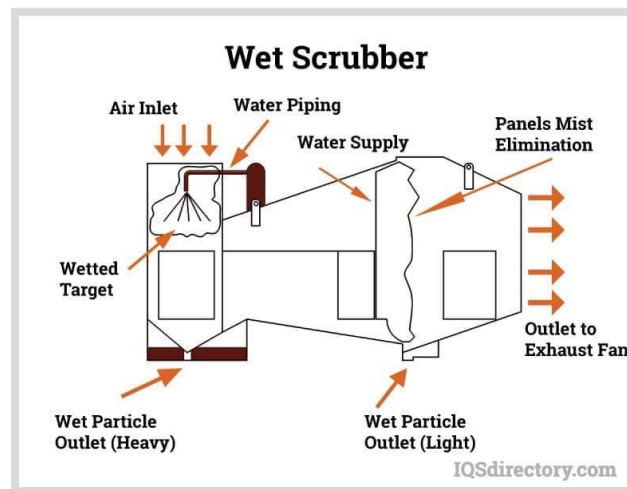
Vertical cartridge dust collecting system replaces bags with cylindrical or oval shaped cartridges that are open on both ends and are lined with pleated filtering media. One end of the cartridge is sealed with the open end used for clean exhaust. The air stream is forced through the outside of the cartridge to the inside. The cartridges are kept clean by compressed air blown into them, which removes built up dust that falls into the hopper below.



- **Wet Scrubber Dust Collection Systems**

A wet scrubber dust collection system uses droplets of water to capture the dust particles. The system has two whirling blades that turn in opposition to each other. The first blade pulls the dust laden air into the mechanism and saturates it with water. As the rotation accelerates, a mist eliminator forces the watered dust particles into the disposal unit. When the dust particles are soaked with water,

they increase in size, which makes them easier to remove. Types of scrubbers include gravity spray towers, cyclone spray chambers, impingement scrubbers, packed bed scrubbers, and venturi scrubbers.

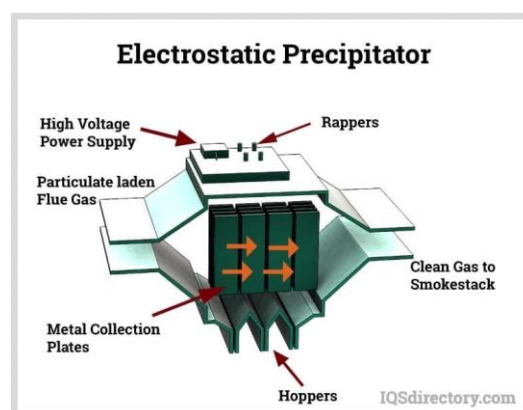


- **Electrostatic Separator System**

Though filtration systems are a popular form of dust collection, there are systems that do not use filters. One of these is an electrostatic precipitator, which uses the force of electrostatic energy to remove particles from the air and has discharge wires and collecting plates. As particles pass through the system, they are ionized and charged, which causes them to be accumulated on a collecting plate. From the collecting plate, the particles are removed by rapping or vibration. The process can be seen in the diagram below.

Electrostatic precipitator dust collectors (ESPS) are effective with large air flow. Common users of ESPS are coal fired power plants and paper mills that have a high volume of gas passing through boilers and process high volumes of air, which is too large for a baghouse system.

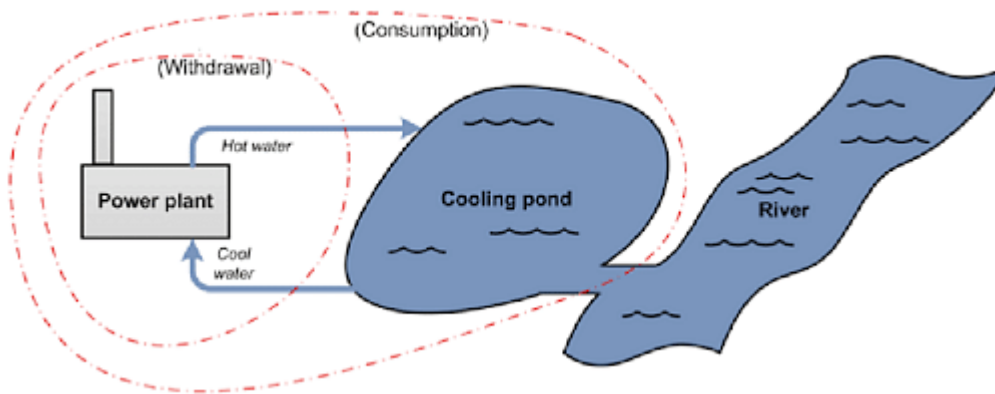
A problem with ESPS systems is how slow particles are to take on an electric charge. For the system to be effective, the particles in the air must be capable of taking on a charge rapidly for quick removal.



Measures to Control Thermal Pollution

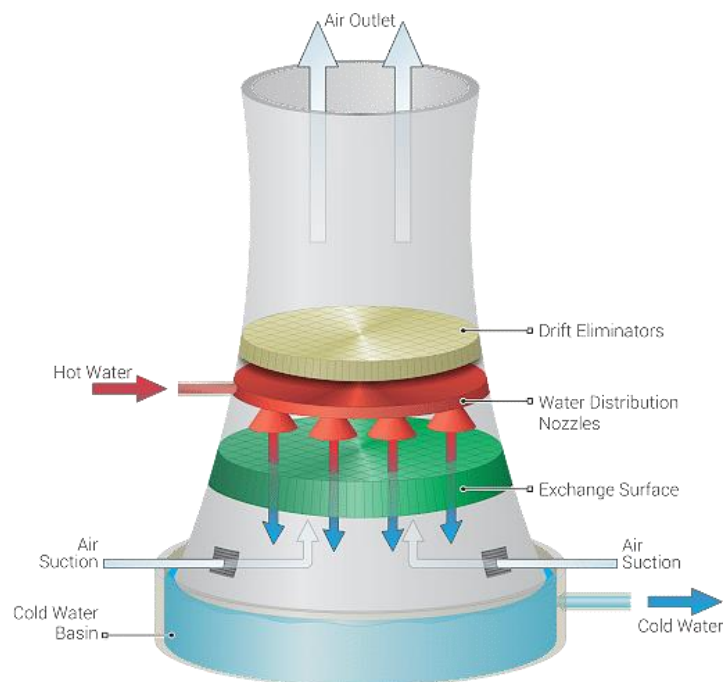
Cooling Ponds

- The most basic ways of controlling thermal discharges are cooling ponds or reservoirs.
- Heated effluents on the surface of water in cooling ponds maximize heat dissipation to the atmosphere while reducing the area and volume of water.
- This is the simplest and most affordable way to chill the water to a very low temperature.
- However, in terms of air-water contact, the method alone is less attractive and inefficient.



Cooling Towers

- The cooling process is defined as the process of taking water from water sources for cooling purposes and then returning it to the water body after passing through the condenser.
- As a result, cooling towers are built to regulate the temperature of water in order to improve the cooling process.
- Cooling towers are primarily used to dissipate recovered waste heat and hence alleviate thermal pollution issues.

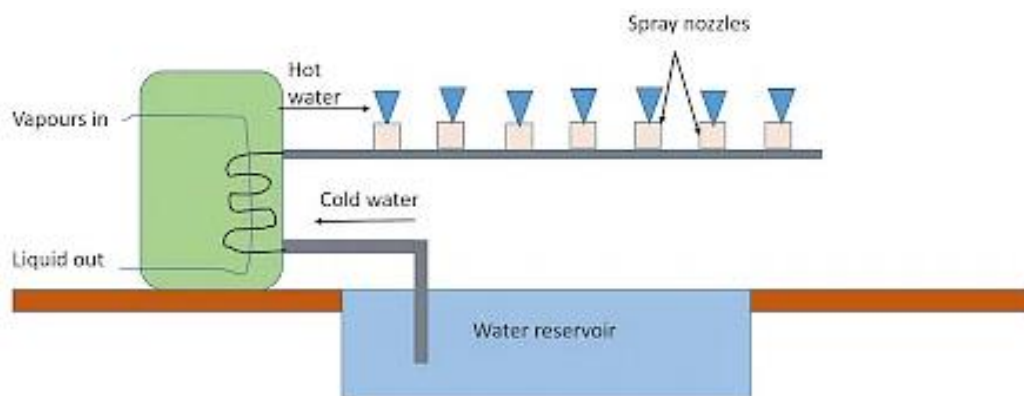


Artificial Lakes

- Artificial lakes are man-made bodies of water that can be used as an alternative.
- It can be created by damming a valley, digging land, or enclosing an area of land with dykes and redirecting a portion of the river flow into the reservoir.
- The heated effluents could be dumped into the lake on one end, and the water could be recovered for cooling on the other.
- Through evaporation, the heat is gradually released. However, these lakes must be replenished on a regular basis.

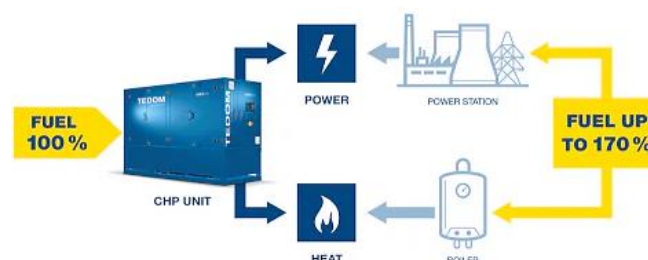
Spray Ponds

- A spray pond is a reservoir where overheated water from a power plant is cooled before reuse by spraying it into the cooler air via nozzles.
- Cooling is accomplished through heat exchange with the surrounding air.
- This includes both conductive heat transfer between the water droplets and the surrounding air as well as evaporative cooling which provides by far the greatest portion, typically 85 to 90 percent, of the total cooling.
- This eventually aids in controlling thermal pollution.



Cogeneration

- Cogeneration is a technique in which the extra thermal energy generated during the generation of electricity is utilized in another manufacturing process that requires that energy.
- It is also known as power district heating and combined heat and power (CHP).
- Waste hot water can be used in industrial and space heating, biological applications such as soil warming, fish culture, livestock shelters, heating greenhouses and homes or other structures near industrial plants.
- This is a common practice in Scandinavian towns and cities and has been proposed for usage in China.



RENOVATION AND MODERNIZATION

New installation is capital-intensive, it is considered prudent to maximize generation from existing power stations to ensure optimal utilization.

Renovation and modernization have a key role to play in ensuring:

- Flexible operation of coal-based power plants with growing competition from renewable energy.
- Catalysing biomass co-firing

	1984–2005	2005–19	2019 onwards (as per the latest draft guidelines)
Primary objective	Generation maximization	Performance optimization and generation maximization	Efficient flexible operation with lower emissions
Primary focus unit	< 200 MW Renovation and modernization after 15 years and life extension after 20 years	> 200 – ≤ 500 Renovation and modernization after 15 years, and life extension after 20 years	> 500 MW Can be done before the stipulated time based on improving flexible generation
Key focus areas	Maximize the generation from existing power stations to ensure optimal utilization of resources, reliability, efficiency and availability	Included specific issues for renovation and modernization to maintain rated capacity and to deal with issues such as deteriorated coal quality and lower plant load factor. Environmental protection was considered, but it was limited to ESP upgradation. Renovation and modernization after 15 years. Life extension after 20 years.	High level of automation to ensure flexible and improved dynamic operations to work in tandem with renewable energy. Renovation and modernization interventions may be needed for refurbishments to improve plant efficiency at part load operation as well. Need for new emissions control equipment installations in power plants for environmental compliance. Biomass utilization for power generation through co-firing in thermal power plants. Conversion of coal-fired plants to biomass power plants. Lowering water consumption in coal-fired power plants. Due to uncertainties in the future operational regime of thermal power generation, life extension of shorter duration may have to be considered.

BIOMASS COFIRING

It has been generally accepted that co-firing biomass with coal can offer a quick, cost-effective way to partially decarbonize power generation in the short to medium term, especially in India, where agro-residue is abundant.

- As per International Energy Agency's (IEA) Roadmap on Biomass Heat and Power, biomass-based power generation will increase by at least a factor of ten from today till 2050, accounting for 7.5 per cent of world's electricity generation. This biomass-based power generation will almost entirely be based on combustion and co-firing technologies.
- The substitution of only 10 per cent of coal in the current globally installed coal-fired electrical capacity would result in installation of about 160–180 GW biomass power capacity, which is 2.5 times
- the current globally installed biomass power capacity. Currently, about 230 power and combined heat and power plants using co-firing techniques are in operation.
- Co-firing also helps to extend the life of a plant.
- Co-firing can play an important transitional role in the decarbonization of the coal fleet.
- CEA has issued an advisory to all thermal power generating plants and utilities to endeavour to use
- 5 to 10 per cent blend of biomass pellets, made primarily from agro-residue, along with coal, after assessing the technical feasibility and safety aspects.
- National Thermal Power Corporation (NTPC) has successfully demonstrated co-firing of 7 per cent blend of biomass pellets with coal at its Dadri power plant. This can be replicated in other coal-fired power plants.
- In September 2019, Ministry of New and Renewable Energy (MNRE) notified that power produced from biomass co-firing in coal-based power plants is renewable energy and will be eligible for non-solar renewable purchase obligation (RPO).
- Thus, if both utility and captive capacity attempts to utilize 5–10 per cent co-firing, it can amount to large-scale agro-residue utilization. Based on the present use of biomass co-firing of 5 to 10 per cent, some 50 to 100 million tonnes of coal will be replaced by biomass by 2030. It is equivalent to reducing 90 to 180 million of CO₂ emissions.

COAL BENEFICIATION

Traditionally, most coal preparation was primarily concerned with sulfur and ash reduction. More sophisticated processes have more narrow and complex physical and chemical requirements for coal stock. High ash content is among the reasons why Indian coal scores poorly on energy value. High ash content (of 30 to 50 per cent) creates many problems for coal users, including difficulty in pulverization, poor emissivity and flame temperature, low radiative transfer, and generation of excessive amounts of fly ash containing large quantities of un-burnt carbon. This reduces the efficiency and increases the auxiliary power consumption of the plant. It is one of the reasons for higher CO₂ emissions from Indian power plants.

Improvements in power plant efficiency through the use of clean (washed) coal can significantly reduce CO₂ emissions. CO₂ emissions can be reduced by 2 to 3 per cent by using 34 per cent ash coal as against 42 per cent ash coal. Coal is treated in processes called “beneficiation” to prepare a material that meets the certain requirements. These processes may result in low ash quantity and low sulphur content. They include the following:

- Sizing, controlled by a crushing and screening process

- increasing heating value, by removing non-combustible ash and rock by gravity separation
- Removing or controlling undesirable mineral and chemical components (sulfur, sodium, and trace elements) by a combination of gravity separation and blending.

Though washing increases the overall cost of the coal, the benefits accrued in terms of savings in transportation, operation and maintenance cost, and efficiency make the process financially sustainable. State-of-the-art technologies such as supercritical and ultra-supercritical pulverized coal combustion or IGCC also benefit from the use of upgraded coals.

Coal Beneficiation is a process where coal is subjected to a medium of defined specific gravity. Heavier material sinks (rejects) and lighter coal floats (clean coal). This specific gravity is defined by the 'washability' characteristics of the coal seam. It varies for different seams in the same area and for different geographical areas. There are two separate aspects to upgrading coal quality. One is the possible short-term benefits that result from using upgraded coals in existing power plant boilers, including reduction in emissions that contribute to local pollution. The other is the longer-term benefit arising from the use of advanced clean coal technologies which may demand upgraded coal by design to realise the potential for increased thermal efficiency. There are other process benefits of coal upgrading, but they are mainly second-order effects. For example, reducing the ash content of coal may make it easier to grind, so that the energy used by mills is reduced. The amount of pyrite, a hard and reactive mineral, present is also likely to be reduced in washed coal.

Inconsistency in the quality of coal supplied is a recurring problem with ROM coal use in India. It is possible that in the case of power plants which draw their supplies from individual mines, there would be greater consistency in the quality of coal. This is so in the case of some of the thermal plants that are linked to specific mines. At times, it was reported that four to five mines were linked to a single power plant depending upon the demand with greater variation in coal quality. Transportation and seasonal variations contributed to the deterioration in quality that was often beyond the tolerance of the power plant.

The ash content in coal, as delivered to power plants in India, currently averages about 40 percent. With few exceptions, the majority of coal-fired power plants are likely to receive coal from more than one source. As most of the plants do not have blending and homogenisation facilities, the multiplicity of supply sources leads to the problems of inconsistency in coal quality.

Benefits of Coal Beneficiation

- REDUCTION IN CARBON DIOXIDE EMISSIONS

CO₂ emissions from power plants depend on the carbon content of the coal and the quantity of air used for combustion. When combustion of coal is incomplete, a small portion of the un-burnt carbon goes with the fly ash and the remaining un-burnt carbon goes in the bottom ash. Improvements in power plant efficiency through the use of clean (washed) coal can have significant benefits in terms of reduction in CO₂ emissions. Test results have demonstrated that CO₂ emissions in the range of 1.11 kilogram per kilowatt hour (kg/kWh) of power generated are reduced by 6.5 percent to 1.045 kg/kWh when using 30 percent ash coal versus 42 percent ash.

- REDUCTION IN OTHER POLLUTANTS

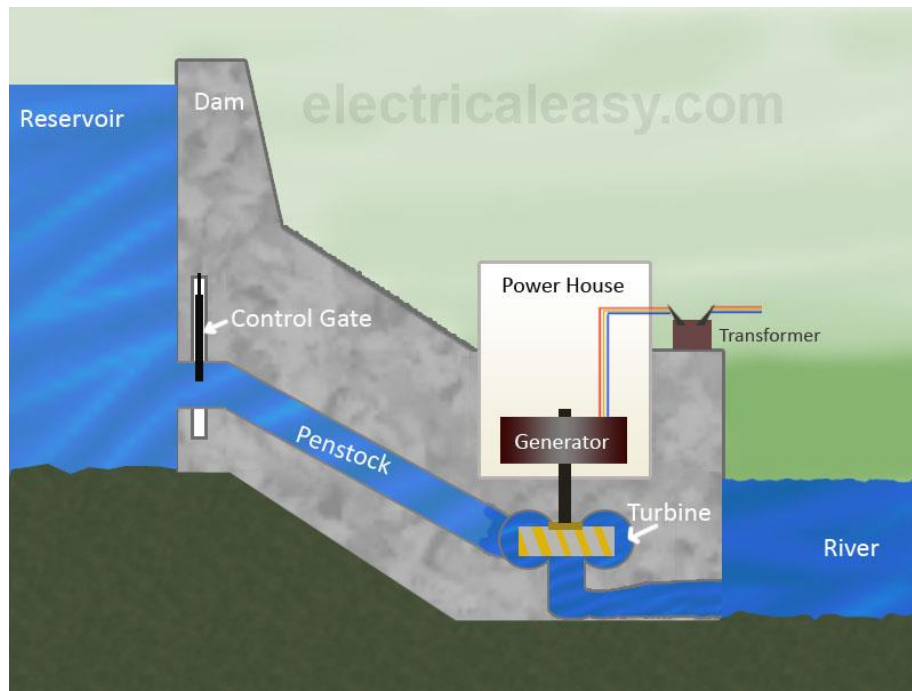
The emission of pollutants like SO_x, NO_x, and suspended particulate matter (SPM) from coal-fired power plants is a source of pollution for surrounding areas unless appropriate control measures are taken. In general, NO_x and SO_x are formed from the combustion of coal where air is used or where nitrogen and sulphur is present in the fuel. SO_x, mainly sulphur dioxide (SO₂), is produced from the

combustion of the sulphur contained in coal of various types. SOX emissions from coal combustion mainly depend on the sulphur content in the coal unlike the emissions of CO₂ and NO_x which depend on the operating conditions (air intake) and the design of the plant. Sulphur content in Indian coal (barring lignite, which has higher sulphur content) is much lower than that of coal from other countries. The range of SO₂ emission for Indian power plants was estimated between 5.210g/kWh-9.899g/kWh.

- REDUCTION IN ASH HANDLING

By reducing the ash content in coal in the pre-combustion stage through washing, not only is the harmful effect of ash reduced, but also the cost of handling ash. Coal washing reduces fly ash generation in the post-combustion stage and extends the life of ash disposal landfills. Using washed coal at a plant would extend a given ash disposal site by 12- 20 percent.

- ECONOMIC BENEFITS OF COAL BENEFICIATION (PLANT OPERATIONS)
 - Thermal power plants have a performance advantage
 - Reduce damage to coal handling equipment such as conveyors, pulverisers, crushers and storage units
 - Directly affects the profitability of a coal plant over the long term
 - Lower ash coal will also result in reduction in freight costs for transporting the same energy content.



The above image shows the typical layout of a **hydroelectric power plant** and its basic components.

Dam and Reservoir: The dam is constructed on a large river in hilly areas to ensure sufficient water storage at height. The dam forms a large reservoir behind it. The height of water level (called as water head) in the reservoir determines how much of potential energy is stored in it.

Control Gate: Water from the reservoir is allowed to flow through the penstock to the turbine. The amount of water which is to be released in the penstock can be controlled by a control gate. When the control gate is fully opened, maximum amount of water is released through the penstock.

Penstock: A penstock is a huge steel pipe which carries water from the reservoir to the turbine. Potential energy of the water is converted into kinetic energy as it flows down through the penstock due to gravity.

Water Turbine: Water from the penstock is taken into the water turbine. The turbine is mechanically coupled to an electric generator. Kinetic energy of the water drives the turbine and consequently the generator gets driven. There are two main types of water turbine; (i) Impulse turbine and (ii) Reaction turbine. Impulse turbines are used for large heads and reaction turbines are used for low and medium heads.

Generator: A generator is mounted in the power house and it is mechanically coupled to the turbine shaft. When the turbine blades are rotated, it drives the generator and electricity is generated which is then stepped up with the help of a transformer for the transmission purpose.

Surge Tank: Surge tanks are usually provided in high or medium head power plants when considerably long penstock is required. A surge tank is a small reservoir or tank which is open at the top. It is fitted between the reservoir and the power house. The water level in the surge tank rises or falls to reduce the pressure swings in the penstock. When there is sudden reduction in load on the turbine, the governor closes the gates of the turbine to reduce the water flow. This causes pressure to increase abnormally in the penstock. This is prevented by using a surge tank, in which the water level rises to reduce the pressure. On the other hand, the surge tank provides excess water needed when the gates are suddenly opened to meet the increased load demand.

Types Of Hydro-Power Plants

Conventional Plants:

Conventional plants use potential energy from dammed water. The energy extracted depends on the volume and head of the water. The difference between height of water level in the reservoir and the water outflow level is called as water head.

Pumped Storage Plant:

In pumped storage plant, a second reservoir is constructed near the water outflow from the turbine. When the demand of electricity is low, the water from lower reservoir is pumped into the upper (main) reservoir. This is to ensure sufficient amount of water available in the main reservoir to fulfil the peak loads.

Run-Of-River Plant:

In this type of facility, no dam is constructed and, hence, reservoir is absent. A portion of river is diverted through a penstock or canal to the turbine. Thus, only the water flowing from the river is available for the generation. And due to absence of reservoir, any oversupply of water is passed unused.

Advantages Of A Hydroelectric Power Plant

- No fuel is required as potential energy is stored water is used for electricity generation
- Neat and clean source of energy
- Very small running charges - as water is available free of cost
- Comparatively less maintenance is required and has longer life
- Serves other purposes too, such as irrigation

Disadvantages

- Very high capital cost due to construction of dam
- High cost of transmission – as hydro plants are located in hilly areas which are quite away from the consumers

ENVIRONMENTAL IMPACTS OF GEOTHERMAL ENERGY

The most widely developed type of geothermal power plant (known as hydrothermal plants) are located near geologic “hot spots” where hot molten rock is close to the earth’s crust and produces hot water. In other regions enhanced geothermal systems (or hot dry rock geothermal), which involve drilling into Earth’s surface to reach deeper geothermal resources, can allow broader access to geothermal energy.

Geothermal plants also differ in terms of the technology they use to convert the resource to electricity (direct steam, flash, or binary) and the type of cooling technology they use (water-cooled and air-cooled). Environmental impacts will differ depending on the conversion and cooling technology used.

Water quality and use

Geothermal power plants can have impacts on both water quality and consumption. Hot water pumped from underground reservoirs often contains high levels of sulfur, salt, and other minerals. Most geothermal facilities have closed-loop water systems, in which extracted water is pumped directly back into the geothermal reservoir after it has been used for heat or electricity production. In such systems, the water is contained within steel well casings cemented to the surrounding rock.

Depending on the cooling technology used, geothermal plants can require between 1,700 and 4,000 gallons of water per megawatt-hour. However, most geothermal plants can use either geothermal fluid or freshwater for cooling; the use of geothermal fluids rather than freshwater clearly reduces the plants overall water impact.

Most geothermal plants re-inject water into the reservoir after it has been used to prevent contamination and land subsidence (see Land Use below). In most cases, however, not all water removed from the reservoir is re-injected because some is lost as steam. In order to maintain a constant volume of water in the reservoir, outside water must be used. The amount of water needed depends on the size of the plant and the technology used; however, because reservoir water is “dirty,” it is often not necessary to use clean water for this purpose.

Air emissions

The distinction between open- and closed-loop systems is important with respect to air emissions. In closed-loop systems, gases removed from the well are not exposed to the atmosphere and are injected back into the ground after giving up their heat, so air emissions are minimal. In contrast, open-loop systems emit hydrogen sulfide, carbon dioxide, ammonia, methane, and boron. Hydrogen sulfide, which has a distinctive “rotten egg” smell, is the most common emission.

Once in the atmosphere, hydrogen sulfide changes into sulfur dioxide (SO₂). This contributes to the formation of small acidic particulates that can be absorbed by the bloodstream and cause heart and lung disease. Sulfur dioxide also causes acid rain, which damages crops, forests, and soils, and acidifies lakes and streams. However, SO₂ emissions from geothermal plants are approximately 30 times lower per megawatt-hour than from coal plants, which is the nation’s largest SO₂ source.

Some geothermal plants also produce small amounts of mercury emissions, which must be mitigated using mercury filter technology. Scrubbers can reduce air emissions, but they produce a watery sludge composed of the captured materials, including sulfur, vanadium, silica compounds, chlorides, arsenic, mercury, nickel, and other heavy metals. This toxic sludge often must be disposed of at hazardous waste sites.

Land use

The amount of land required by a geothermal plant varies depending on the properties of the resource reservoir, the amount of power capacity, the type of energy conversion system, the type of cooling system, the arrangement of wells and piping systems, and the substation and auxiliary building needs. The Geysers, the largest geothermal plant in the world, has a capacity of approximately 1,517 megawatts and the area of the plant is approximately 78 square kilometers, which translates to approximately 13 acres per megawatt. Like the Geysers, many geothermal sites are located in remote and sensitive ecological areas, so project developers must take this into account in their planning processes.

Land subsidence, a phenomenon in which the land surface sinks, is sometimes caused by the removal of water from geothermal reservoirs. Most geothermal facilities address this risk by re-injecting wastewater back into geothermal reservoirs after the water's heat has been captured.

Hydrothermal plants are sited on geological "hot spots," which tend to have higher levels of earthquake risk. There is evidence that hydrothermal plants can lead to an even greater earthquake frequency. Enhanced geothermal systems (hot dry rock) can also increase the risk of small earthquakes. In this process, water is pumped at high pressures to fracture underground hot rock reservoirs similar to technology used in natural gas hydraulic fracturing. Earthquake risk associated with enhanced geothermal systems can be minimized by siting plants an appropriate distance away from major fault lines. When a geothermal system is sited near a heavily populated area, constant monitoring and transparent communication with local communities is also necessary.

Life-cycle global warming emissions

In open-loop geothermal systems, approximately 10 percent of the air emissions are carbon dioxide, and a smaller number of emissions are methane, a more potent global warming gas. Estimates of global warming emissions for open-loop systems are approximately 0.1 pounds of carbon dioxide equivalent per kilowatt-hour. In closed-loop systems, these gases are not released into the atmosphere, but there are still some emissions associated with plant construction and surrounding infrastructure.

Enhanced geothermal systems, which require energy to drill and pump water into hot rock reservoirs, have life-cycle global warming emission of approximately 0.2 pounds of carbon dioxide equivalent per kilowatt-hour. To put this into context, estimates of life-cycle global warming emissions for natural gas generated electricity are between 0.6 and 2 pounds of carbon dioxide equivalent per kilowatt-hour and estimates for coal-generated electricity are 1.4 and 3.6 pounds of carbon dioxide equivalent per kilowatt-hour.

ENVIRONMENTAL IMPACTS OF NUCLEAR ENERGY

The use of nuclear energy presents an interesting dilemma. On the one hand, nuclear electricity produces no carbon emissions, a major sustainable advantage in a world facing climate change. On the other hand, there is environmental risk of storing spent fuel for thousands or hundreds of thousands of years, societal risk of nuclear proliferation, and the impact of accidental releases of radiation from operating reactors. Thoughtful scientists, policy makers, and citizens must weigh these advantages and disadvantages.

Advantages

Disadvantages

Despite its benefits, nuclear power has downsides. It requires more water than any other energy source. Water used for cooling is released back into the environment, and while it does not contain radioactive materials or other harmful chemicals, it is warmer than before. This is called thermal pollution, and it can harm aquatic life, which are adapted to cooler temperatures. Surface mining for uranium ore degrades habitat and releases toxins from underground (similar to surface mining for coal). Nuclear power plants are expensive to build and maintain, and they require large amounts of metal and concrete. While nuclear accidents are rare, they can cause great harm, and their impacts are long-lasting. Furthermore, the problem of safely disposing spent nuclear fuel remains unresolved. The latter two concerns are discussed in more detail below.

Nuclear Wastes: The main environmental challenge for nuclear power is the **wastes** including high-level radioactive waste, low-level radioactive waste, and uranium mill tailings. These materials have long radioactive half-lives and thus remain a threat to human health for thousands of years.

High-level radioactive waste (HLRW) consists of used nuclear reactor fuel (spent nuclear fuel rods). These contain the products of nuclear fission, which are radioactive themselves. This HLRW is temporarily stored in a pool at the nuclear power plant or a dry cask, steel cylinders within another container, made of steel or concrete. Dry casks contain inert (nonreactive) gas and may be located at the power plant, a decommissioned power plant, or a separate storage site. High-level radioactive waste may only be moved to a dry cask after one year of cooling in a pool.

Some countries reprocess (recycle) spent nuclear fuel, but no recycling or reprocessing facility or a federal waste repository is currently licensed in the United States. Reprocessing separates the useable fraction of spent fuel and recycles it through the reactor, using a greater fraction of its energy content for electricity production, and sends the remaining high-level waste to permanent geologic storage.

The primary motivation for reprocessing is greater use of fuel resources, extracting about 25 percent more energy than the once through cycle. A secondary motivation for recycling is a significant reduction of the permanent geologic storage space (to 20% or less of what would otherwise be needed) and time (from hundreds of thousands of years to thousands of years). While these advantages seem natural and appealing from a sustainability perspective, they are complicated by the risk of theft of nuclear material from the reprocessing cycle for use in illicit weapons production or other non-sustainable ends. At present, France, the United Kingdom, Russia, Japan, and China engage in some form of reprocessing; the United States, Sweden, and Finland do not reprocess.

Low-level radioactive waste (LLRW) refers to items that were exposed to radiation includes clothing, filters, and gloves. These can be contained with concrete or lead. Low-level waste is typically stored at the nuclear power plant, either until it has decayed away and can be disposed of as ordinary trash, or until amounts are large enough for shipment to LLRW disposal sites.

Enrichment of uranium produces depleted uranium hexafluoride (DUF6), or *uranium mill tailings*, as a byproduct, which does not have high enough concentrations of ^{235}U to use as nuclear fuel but is still hazardous. Tailings represent the greatest percentage of nuclear waste by volume, and there are more than 200 million tons of radioactive mill-tailings in the United States. Tailings contain several radioactive elements including radium, which decays to produce radon, a radioactive gas. They are stored in impoundments, lined pits in the ground that are flooded with water, in remote areas. Deconversion involves chemically treating the tailings to reduce their hazards so that they can be stored as LLRW.

Nuclear Disasters: There are many other regulatory precautions governing permitting, construction, operation, and decommissioning of nuclear power plants due to risks from an uncontrolled nuclear reaction. The potential for contamination of air, water and food is high should an uncontrolled reaction occur. Even when planning for worst-case scenarios, there are always risks of unexpected events. The nuclear accidents at Three Mile Island, Chernobyl, and Fukushima raised concerns about the safety of nuclear power.